Burak Bulut, Ph.D.

Bioinformatics Scientist & Computational Biologist

Wiesbaden, Germany • Email • LinkedIn • GitHub • Google Scholar

PROFESSIONAL SUMMARY

Bioinformatics Scientist and Computational Biologist (Ph.D.) with 6+ years of experience in genomics, transcriptomics, NGS data analysis, and workflow automation across research environments. Proven track record in designing multi-year studies, engineering reproducible bioinformatics pipelines (Snakemake, Docker, SLURM) on Linux/HPC clusters, and leading scientific coordination including DFG grant proposal development. Author of 7 scientific publications (6 peer-reviewed articles, 1 preprint; 4 first-author), combining strong wet-lab experimental grounding with computational prototyping in Python/R, local LLM fine-tuning, and RAG-enabled analysis workflows.

CORE TECHNICAL COMPETENCIES

NGS & Genomics	Whole-Genome Sequencing (WGS), RNA-Seq, Variant & SNP Calling (GATK, BWA-MEM, Picard, SAMtools), Differential Gene Expression (DESeq2, featureCounts, Bioconductor), QC & Preprocessing (FastQC, MultiQC, Trimmomatic), *de novo* Mutation Rate & Stress-Adaptation Genomics
Programming & Pipelines	Python, R (tidyverse, ggplot2, Bioconductor), Bash/Shell Scripting, Snakemake, Docker, Git, JupyterBook, Reproducible Workflow Engineering
HPC & Infrastructure	Linux/Ubuntu Server Administration, High-Performance Computing (HPC) Cluster Management, SLURM Workload Manager, AWS (Basic)
AI & Data Science	AI-Assisted Bioinformatics Prototyping, Local LLM Fine-Tuning, Retrieval-Augmented Generation (RAG) Systems, Prompt Engineering, Advanced Statistical Modeling
Databases & Wet-Lab	NCBI, Ensembl, KEGG, UniProt, GO • DNA/RNA Extraction, PCR, Gel Electrophoresis, *in vivo* ROS Quantification, Fluorescence Microscopy
Languages	Turkish (Native) • English (Fluent / Full Professional Proficiency) • German (A2 – Basic Working Proficiency)

PROFESSIONAL EXPERIENCE

Independent Researcher & AI/Bioinformatics Developer — Jul 2025 – Present

Wiesbaden / Frankfurt Rhine-Main Metropolitan Region, Germany

- **Grant & Scientific Coordination:** Led the end-to-end development of a Deutsche Forschungsgemeinschaft (DFG) research grant proposal, taking primary responsibility for experimental planning, scientific coordination, and technical framing.
- **AI-Enabled Bioinformatics Prototyping:** Architected prototype agentic and RAG-enabled bioinformatics workflows utilizing local LLM fine-tuning and structured prompt engineering for automated QC reasoning, tool orchestration, and biological interpretation.
- **Doctoral Completion & Publishing:** Finalized doctoral dissertation and spearheaded multiple first-author and collaborative peer-reviewed publications across evolutionary genomics and stress transcriptomics.

Computational Biologist | NGS Data Scientist — Jul 2022 – Jul 2025

Senckenberg Biodiversity and Climate Research Centre (SBiK-F), Frankfurt am Main, Germany

- **Pipeline Engineering & Automation:** Engineered, automated, and maintained scalable end-to-end WGS and RNA-Seq pipelines using Snakemake, Python, and Bash (integrating BWA-MEM, Picard, SAMtools, GATK, and DESeq2) on Linux/SLURM HPC clusters.
- **Large-Scale Multi-Omics Analysis:** Processed and analyzed high-throughput genomic and transcriptomic datasets to quantify germline mutation frequencies, characterize cryptic genetic variation, and evaluate transcriptomic responses to environmental stressors.
- **Experimental & Study Design:** Designed and executed complex, multi-year multigenerational mutation accumulation and ecotoxicological exposure experiments, integrating wet-lab assays (*in vivo* ROS measurement, fluorescence microscopy) with computational evaluations.
- **Statistical Modeling & Reporting:** Performed rigorous statistical evaluations and generated publication-grade visualizations and actionable technical reports using R (tidyverse, ggplot2) and Python for interdisciplinary teams.
- **Teaching & Mentorship:** Taught molecular ecology coursework over 3 academic years, trained researchers in Unix/HPC bioinformatics workflows, and supervised a B.Sc. thesis project to successful completion.

SELECTED TECHNICAL PROJECTS & OPEN-SOURCE TOOLS

PipeWatch (HPC Pipeline & Resource Monitor): Python-based command-line tool for real-time monitoring of bioinformatics pipeline execution, job states, and compute resource utilization on Linux/SLURM HPC systems.

AI Bioinformatics Agent Prototype: Modular local LLM and RAG-based agent workflows designed for automated quality-control reasoning, CLI tool use, and structured biological data interpretation.

EDUCATION

Ph.D. in Molecular Ecology & Bioinformatics — 2026

Johannes Gutenberg University Mainz & Senckenberg Biodiversity and Climate Research Centre, Germany

Dissertation: *Deconstructing the proximate drivers of the germline mutation rate in Chironomus riparius* (OpenScience Mainz)

M.Sc. in Biology (Specialization: Bioinformatics) **2022**

Bolu Abant İzzet Baysal University, Bolu, Turkey — Thesis: Meta-analysis of RNA-Seq data for sugar beet (Beta vulgaris L.) under biotic and abiotic stresses

B.Sc. in Biology **2019**

Bolu Abant İzzet Baysal University, Bolu, Turkey

PUBLICATIONS

PEER-REVIEWED JOURNAL ARTICLES

1. **Bulut, B.**, & Pfenninger, M. (2026). Developmental speed and chronological time exert opposing effects on the spontaneous mutation rate in *Chironomus riparius*. *Evolution Letters*, 10(3), 283–293. DOI
2. Pfenninger, M., Nieto-Blazquez, M. E., & **Bulut, B.** (2026). Divergent Thermal Reaction Norms of the Spontaneous Mutation Rate Among Populations of *Chironomus riparius*. *Molecular Ecology*, 35(15), e70503. DOI
3. **Bulut, B.**, Geiss, M., Bernard, M., Doria, H. B., Feldmeyer, B., & Pfenninger, M. (2025). Dissecting Oxidative Stress and Organismic Response to a Temperature Gradient in the Midge *Chironomus riparius*. *Ecology and Evolution*, 15(12), e72625. DOI
4. Eberhardt, L., Doria, H. B., **Bulut, B.**, Feldmeyer, B., & Pfenninger, M. (2025). Transcriptomics predicts artificial light at night's (ALAN) negative fitness effects and altered gene expression patterns in the midge *Chironomus riparius* (Diptera: Chironomidae). *Environmental Pollution*, 369, 125827. DOI
5. **Bulut, B.**, Rigano, L., Doria, H. B., Gemüth, G., & Pfenninger, M. (2024). A multigenerational study can detect the evolutionary response to BaP exposure in the non-biting freshwater midge *Chironomus riparius*. *Chemosphere*, 358, 142242. DOI
6. **Bulut, B.**, Gürel, S., Ünüvar, Ö. C., Gürel, E., Şahin, Y., Çabuk, U., & Ünlü, E. S. (2023). Meta Analysis of Sugar Beet (*Beta vulgaris L.*) Transcriptome Profiles Under Different Biotic and Abiotic Stress Conditions. *Tropical Plant Biology*, 16(3), 199–207. DOI

PREPRINTS

1. Pfenninger, M., Nieto-Blazquez, M. E., Schell, T., Geiss, M., **Bulut, B.**, Eberhardt, L., & Feldmeyer, B. (2025). Pervasive and dynamic release of Cryptic Genetic Variation in *Chironomus riparius*: Rethinking adaptation in fluctuating environments. *bioRxiv*. DOI